BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SUKKUR

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2019

CHEMISTRY

Past Paper 2019 • Paper Code: A

Retyped faithfully from the original board paper — same questions, no changes.

Section - A (Multiple Choice Questions)

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** A

Note: Q.16: Choose the correct answer from the given options and write the same in your answer copy. Note: Cutting and Overwriting is not allowed.

Q1. The solution that contains greater amount of solute as compared to the amount of solvent is known as _____

(A) Neutral solution
(B) Dilute solution
(C) Concentrated solution
(D) Weak solution

Q2. The process in which molecules escape from the surface of liquid is called _____

(A) Sublimation
(B) Evaporation
(C) Boiling
(D) Melting

Q3. The optimum condition of temperature for the maximum yield of SO3 in contact process is _____

(A) 250 C°
(B) 350 C°
(C) 550 C°
(D) 450 C°

Q4. Natural gas mainly consists of _____

(A) Propane
(B) Methane
(C) Ethane
(D) Butane

Q5. The polymer which is used in the manufacturing of bags for shopping purpose is _____

(A) Poly vinyl chloride
(B) Poly ethylene
(C) Epoxy resin
(D) Araldite

Q6. If enthalpy of products is greater than the enthalpy of reactants, then the sign of ΔH will be _____

(A) −
(B) +
(C) ±
(D) None of these

Q7. The deficiency of this leads to the enlargement of thyroid gland

(A) Fluorine
(B) Iodine
(C) Bromine
(D) Astatine

Q8. The chemical formula of the chief ore of Aluminium is _____

(A) Al2O3.2H2O
(B) Al2O3
(C) Al2O3.H2O
(D) Na3AlF6

Q9. Sulphur burns in air with blue flame to produce _____

(A) SO3
(B) H2S
(C) S2Cl2
(D) SO2

Q10. The bond which is formed by one sided sharing of pair of electrons is called _____

(A) Electrovalent bond
(B) Covalent bond
(C) Co-ordinate covalent bond
(D) Ionic bond

Q11. The boiling point of liquid oxygen is _____

(A) -196°C
(B) -182°C
(C) -183°C
(D) -181°C

Q12. Alpha rays have _____ charge

(A) Positive
(B) Negative
(C) Negative & positive
(D) None of these

Q13. Chromite is found in Pakistan at _____

(A) Sindh
(B) Punjab
(C) Baluchistan
(D) Azad Kashmir

Q14. A substance that absorbs moisture from the atmosphere is called _____

(A) Hydroscopic
(B) Amphoteric
(C) Hygroscopic
(D) None of these

Q15. _____ is a dark brown volatile liquid.

(A) Bromine
(B) Chlorine
(C) Fluorine
(D) Iodine

Q16. The quantity of electric charge when one ampere of current is passed for one second is called _____

(A) Faraday
(B) Coulomb
(C) Volt
(D) Watt

Q17. The maximum density of water at 4°C is _____

(A) 0.998 g/cm3
(B) 1.1 g/cm3
(C) 0.918 g/cm3
(D) 1.0 g/cm3

Section - A (Short Answer Questions)

Time Allowed: 2 Hours and 40 minutes | **Maximum Marks:** 40

Note: NOTE:- ATTEMPT ANY EIGHT (08) QUESTIONS. EACH QUESTION CARRIES (5) MARKS.

Q1. Describe the extraction of Aluminum metals from its bauxite ore?

ایلومینیم دھات کو اس کی کچ دھات باکسائٹ سے حاصل کرنے کا عمل بیان کریں۔

Q2. What happens when electric current is passed through acidulated water? Give the reactions at electrodes and mention the products at cathode and anode.

تیزابی پانی میں سے بجلی گزارنے پر کیا ہوتا ہے؟ دونوں الیکٹروڈز پر ہونے والے عمل اور کیتھوڈ اور اینوڈ پر حاصل ہونے والی مصنوعات کا ذکر کریں۔

Q3. Compare the general properties of graphite and diamond and mention two uses of each graphite and diamond.

ڈائمنڈ اور گریفائٹ کی عام خصوصیات کا موازنہ کریں اور ڈائمنڈ اور گریفائٹ کے دو دو استعمال بیان کریں۔

Q4. How are flowers of Sulphur formed?

گندھک کے پھول کیسے بنتے ہیں؟

Q5. Under what conditions are following formed?

(a) Polar covalent bond

(b) Non polar covalent bond

(c) Ionic bond

Q6. Write the balanced equation for the complete combustion of:

مکمل احتراق کے لیے کیمیائی متوازن مساوات لکھیں۔

(a) Methane

(b) Ethene

(c) Acetylene

Q7. What are the lanthanides and actinides? Are they "d" or "f" block elements?

لینتھینائیڈز اور ایکٹینائیڈز کیا ہیں؟ کیا یہ 'd' یا 'f' بلاک کے عناصر ہیں؟

Q8. In what way do isotopes of an element differ from each other?

ایک ہی عنصر کے ہمجا (آئسوٹوپس) ایک دوسرے سے کس طرح مختلف ہوتے ہیں؟

Q9. Why Ozone is important in the upper atmosphere?

بالائی فضا میں اوزون کی کیا اہمیت ہے؟

Q10. The atomic masses of 17Cl35 (75%) and 17Cl37 (25%) calculate the average atomic mass of chlorine?

Q11. Matter exists in three states (solid, liquid and gas) on mixing they produce nine (9) types of solutions. Write these with examples.

Q12. Write note on heavy water and its uses also.

بھاری پانی پر نوٹ لکھیں اور بھاری پانی کے استعمالات بھی بیان کریں۔

Section - B (Long Answer Questions)

Maximum Marks: 28

> Note: NOTE: ATTEMPT ANY TWO (02) QUESTIONS. QUESTION NO: FIFTEEN (15) IS COMPULSORY.

Q13. Define acid-base neutralization? Write in detail the steps for carrying out titration.

ایسڈ۔بیس نیوٹرلائزیشن کی تعریف کریں اور ٹائٹریشن کے عمل کے مراحل تفصیل سے بیان کریں۔

Q14.

(a) Describe and explain how sulphur is extracted from underground deposits by Frasch process?

فراش طریقے سے زیر زمین ذخائر سے گندھک کیسے نکالی جاتی ہے؟

(b) Write down the uses of chlorine.

کلورین کے استعمالات لکھیں۔

Q15. Complete and balance the following equations.

(i) C2H5–OH [illegible in source scan]

(ii) [illegible in source scan] + Cl2 + H2O → [illegible in source scan]

(iii) PbO + HNO3 → [illegible in source scan]

(iv) SO3 + H2SO4 (Conc) → [illegible in source scan]

(v) [illegible in source scan] + HCl → [illegible in source scan]

(vi) K2Cr2O7 + HCl → [illegible in source scan]

(vii) C6H6 + HNO3 → [illegible in source scan]